

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
```

```
(X_train,y_train),(X_test,y_test) = keras.datasets.mnist.load_data()
```

Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
11490434/11490434 ————— 0s 0us/step

```
X_train = X_train.reshape(-1,784)
X_test = X_test.reshape(-1,784)
```

```
X_train = X_train.astype("float32")/255.0
X_test = X_test.astype("float32")/255.0
print(f'Train: {X_train.shape} Test: {X_test.shape}')
```

Train: (60000, 784) Test: (10000, 784)

```
model = keras.Sequential([
keras.Input(shape=(784,)),

layers.Dense(234, activation='relu'),
layers.Dropout(0.2),

layers.Dense(64, activation='relu'),
layers.Dropout(0.2),

layers.Dense(10, activation='softmax'),
])
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 234)	183,690
dropout (Dropout)	(None, 234)	0
dense_1 (Dense)	(None, 64)	15,040
dropout_1 (Dropout)	(None, 64)	0
dense_2 (Dense)	(None, 10)	650

Total params: 199,380 (778.83 KB)

Trainable params: 199,380 (778.83 KB)

Non-trainable params: 0 (0.00 KB)

```
model.compile(
optimizer='adam',
loss='sparse_categorical_crossentropy',
metrics=['accuracy'],
)
```

```
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
```

```
early_stop = EarlyStopping(monitor='val_loss', patience=5,
restore_best_weights=True, verbose=1)
```

```
# Halve learning rate if val loss stalls for 3 epochs
```

```
# history.history is a dict: {'loss':[], 'accuracy':[], 'val_loss':[], 'val_accuracy':[]}
```

Restoring model weights from the end of the best epoch: 7.

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